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[5252]-163

SE (Computer) (I Sem.) EXAMINATION, 2017
DIGITAL ELECTRONICS AND LOGIC DESIGN
(2012 PATTERN)

Time : Two Hours

Maximum Marks : 50

N.B. :— (i) Solve Q. No. 1 or Q. No. 2, Q. No. 3 or Q. No. 4,
Q. No. 5 or Q. No. 6 and Q. No. 7 or Q. No. 8.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Use of Mollier charts, electronic pocket calculator and steam tables are allowed.

(v) Assume suitable data, if necessary.

1. (a) Minimize the following function using K-map and realize using logic gates : [4]

$$F(A, B, C, D) = \sum m(1, 5, 7, 13, 15) + d(0, 6, 12, 14).$$

(b) Convert the following : [2]

$$(175)_{10} = (?)_8$$

(c) List the differences between CMOS and TTL. [6]

Or

2. (a) Convert the following numbers into binary numbers : [4]

(i) $(37)_8$

(ii) $(25.5)_{10}$

(b) Explain the standard TTL characteristics in detail. [6]

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- (c) Represent the following signed number in 2's complement method : [2]
- (i) +17
- (ii) -17.
3. (a) Explain rules for BCD addition with suitable example and design a single digit BCD adder using IC 7483. [6]
- (b) Design a MOD-6 synchronous counter using J-K flip-flops. [6]
- Or*
4. (a) Design a sequence generator using J-K flip-flop sequence is : [6]
- $1 \rightarrow 3 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 1.$
- (b) Design a circuit to convert 4-bit binary to its equivalent gray code. [6]
5. (a) What is ASM chart ? Give its application and explain the MUX controller method with suitable example. [7]
- (b) Write VHDL code for 4-bit adder using structural modelling style. [6]
- Or*
6. (a) Draw the ASM chart for the following state machine. A 2-bit up counter is to be designed with output $Q_A Q_B$ and enable signal 'X'. If $X = 0$, then counter changes the state as 00-01-10-11-00. If $X = 1$, then counter should remain in current state. Design the circuit using J-K flip-flop and suitable MUX. [7]
- (b) Write a VHDL code for 8 : 1 MUX using Behavioural modeling. [6]

7. (a) Draw and explain basic architecture of FPGA in detail. [6]
(b) A combinational circuit is defined by the functions : [7]

$$f_1(A, B, C) = \Sigma m(3, 5, 7)$$

$$f_2(A, B, C) = \Sigma m(4, 5, 7)$$

Implement the circuit with PLA having 3 input and 3 product term with 2 output.

Or

8. (a) Implement 4 : 1 MUX using PAL. [6]
(b) Implement the following functions using PLA : [7]

$$f_1(A, B, C) = \Sigma m(0, 3, 4, 7)$$

$$f_2(A, B, C) = \Sigma m(1, 2, 5, 7)$$